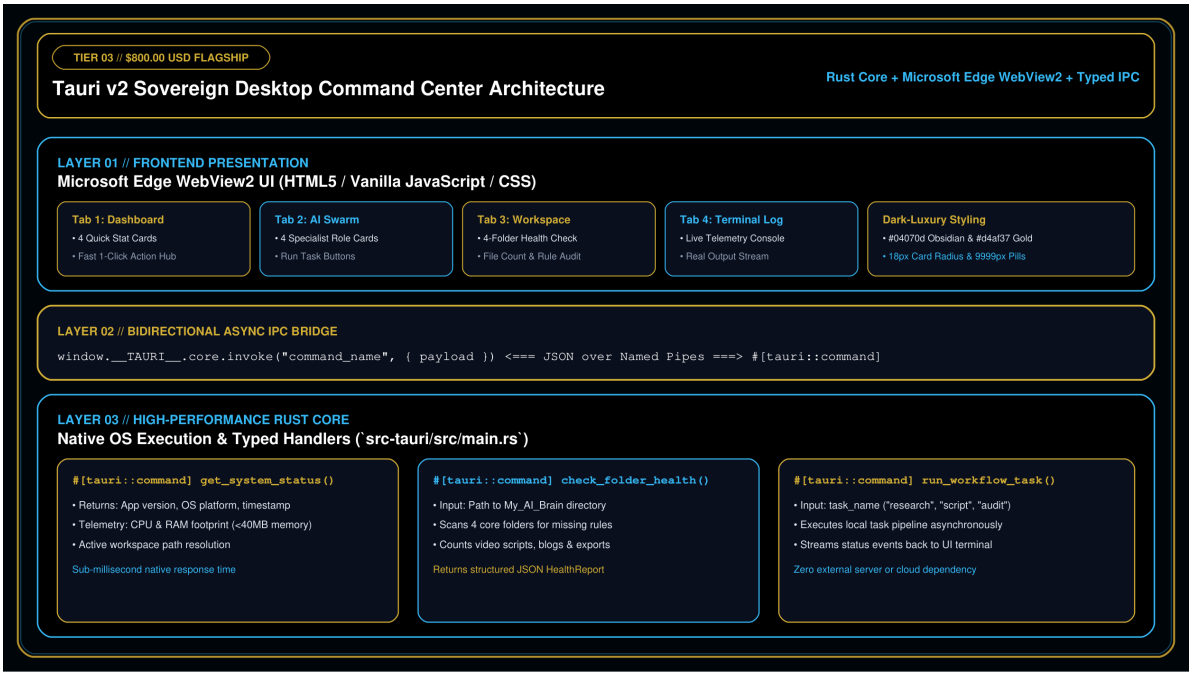
PDF-10: Tauri v2 Desktop App Toolchain & Project Scaffold

Tier 03 Step 1: Rust, Node.js, Microsoft Edge WebView2 & Clean-Room Project Extraction (Page 1/2)

1. WHY TAURI V2 BEATS HEAVY ELECTRON APPS

Legacy desktop web wrappers like Electron bundle an entire Chromium browser into every app, consuming 400MB+ of RAM and producing massive 150MB installers. **Tauri v2** uses your operating system's native **Microsoft Edge WebView2** paired with a high-performance **Rust core**, producing single standalone executables that start in milliseconds and consume under **40MB of RAM**.

2. TAURI V2 ARCHITECTURE OVERVIEW



Frontend: WebView2 UI
Vanilla JavaScript, HTML5, and CSS tokens (#04070d, #d4af37, #38bdf8). Zero heavy node runtime in production.

IPC Bridge: window.__TAURI__
Type-safe asynchronous JSON serialization between web UI and native Rust commands.

Backend: Rust Core (main.rs)
High-speed native file system scanning, subprocess execution, and system telemetry.

Standalone Binary (.exe)
Compiled single-file executable requiring zero Python, Node, or runtime dependencies on user PCs.

→ See Page 2 for the Toolchain Installation & Workspace QuickStart!

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3. PREREQUISITE TOOLCHAIN INSTALLATION (5 MINUTES)

1. **Install Rust & Cargo:** Visit **rust-lang.org** and run *rustup-init.exe*. Choose Default Installation (MSVC).
2. **Visual Studio C++ Build Tools:** In Visual Studio Installer, check *"Desktop development with C++"*.
3. **Node.js LTS:** Install Node.js v18+ or v20+ from **nodejs.org**.
4. **WebView2:** Pre-installed on Windows 10/11. (If needed, download Evergreen Bootstrapper).

4. EXTRACTING & RUNNING YOUR CLEAN-ROOM TAURI V2 STARTER

Unzip **Keystone_Tier03_Sovereign_Workstation_Installer_Suite.zip** and open **tauri_command_center_starter**/:

```
TERMINAL COMMANDS (INSIDE tauri_command_center_starter/):
# 1. Install Tauri CLI dependencies
npm install

# 2. Launch your native desktop command center in live development mode
npm run tauri dev
```

Success Affirmation: A native dark-luxury Windows window opens instantly with your 5 Command Center tabs and live system telemetry. Next, open **PDF-11** to wire custom Rust IPC commands!